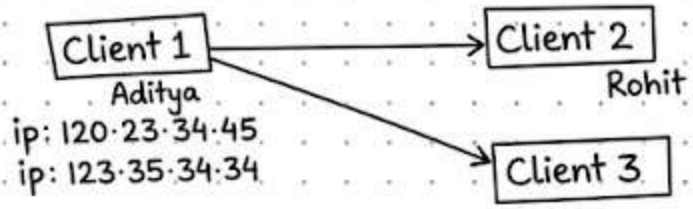
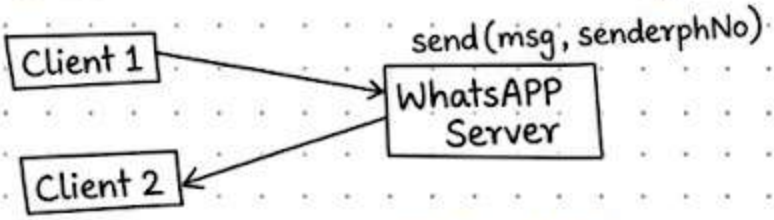
Lecture 17: Designing WhatsAPP | API Gateway | Concluding HLD

- Peer-to-peer communication

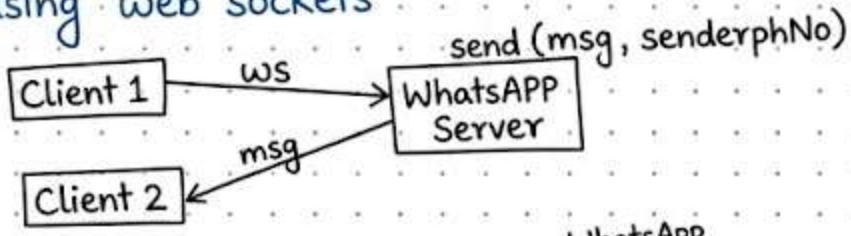


- Client - Server Architecture using HTTP



- Again, client can poll for MSG

- using web sockets



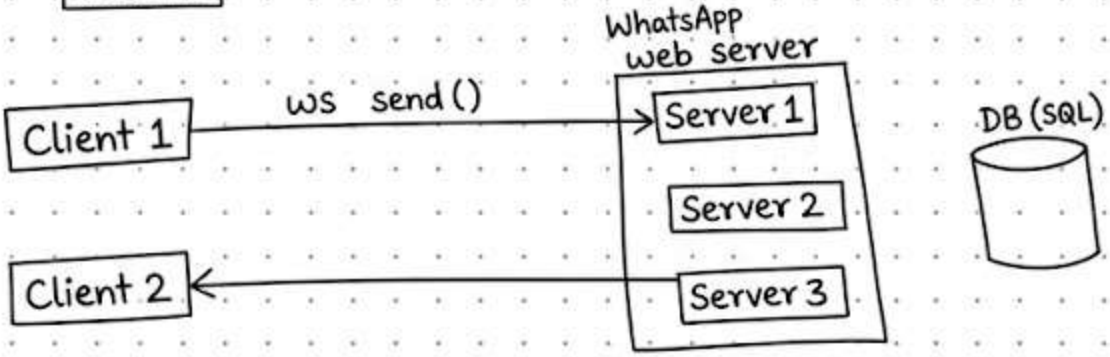
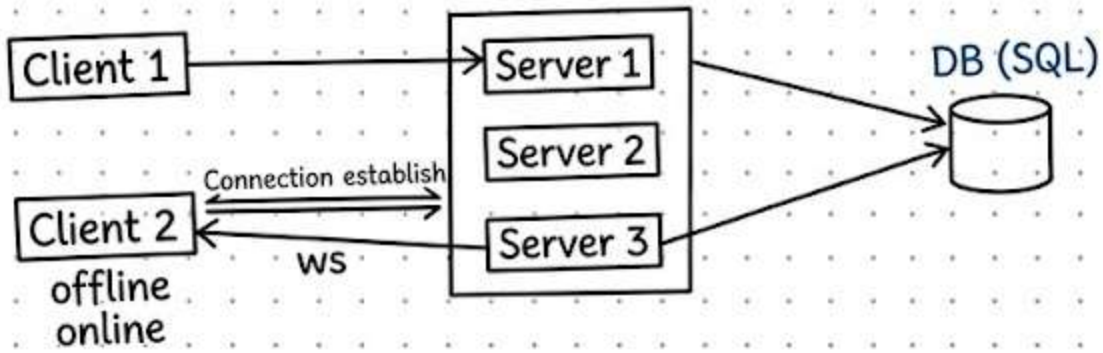


Table: communicate
sender, Reciever, Msg, timestamp

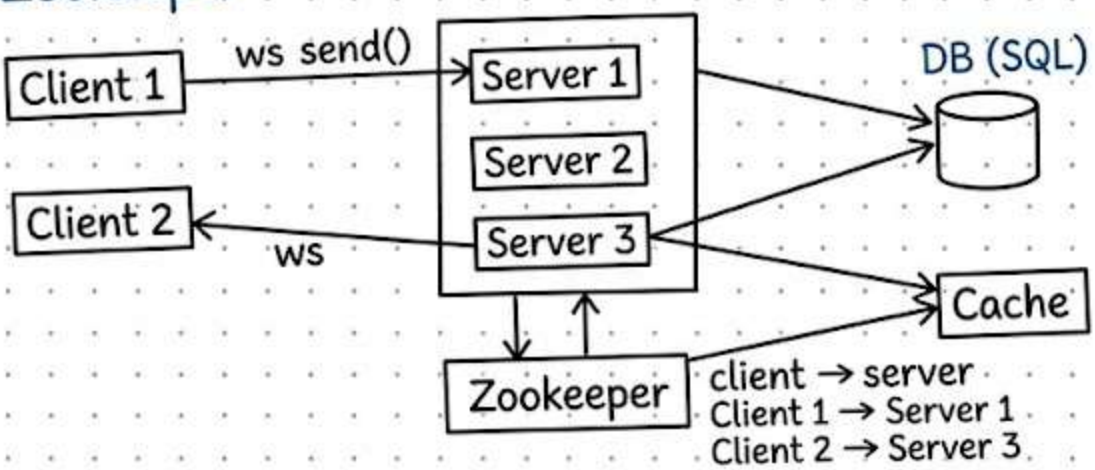
Case I - Both clients are online

Same diagram as previous

Case II - Client 2 is offline



↳ Zookeeper

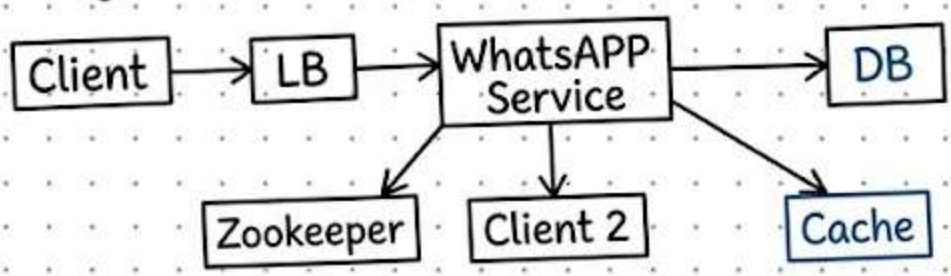


↳ Group chat

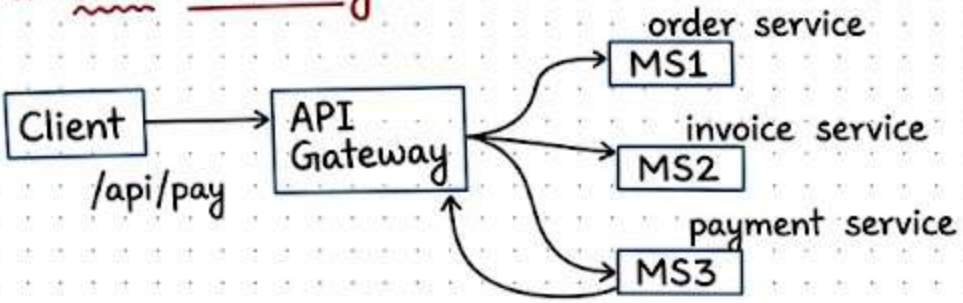
Zookeeper can hold one more mapping of

G_1 → Server

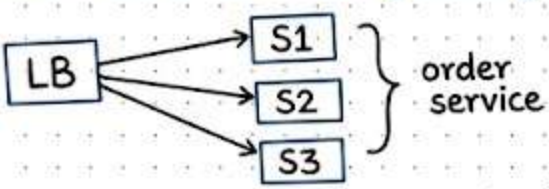
g_1 → C1, C2, C3 , S1, S2, S3



API Gateway

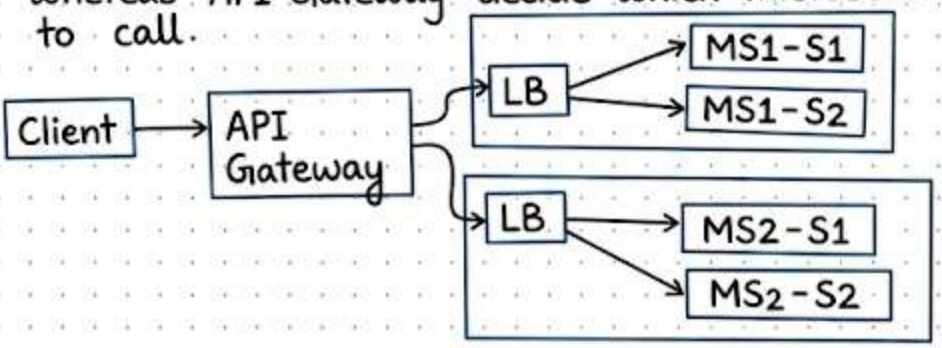


LoadBalancer vs API gateway



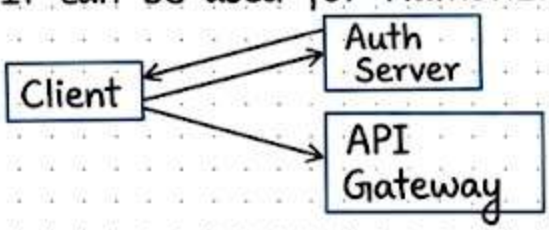
LoadBalancer distributes traffic to multiple instances of same server.

whereas API Gateway decide which microservice to call.



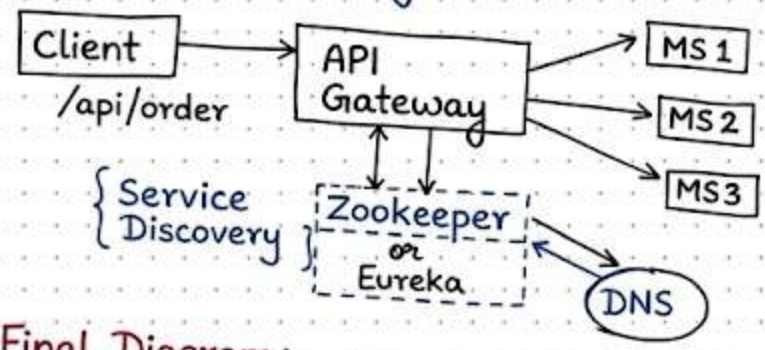
4 What API Gateway do?

1. Transfer API calls to correct microservice.
2. It can be used for Authorization / Authentication



We can Implement Rate limiting logic to API Gateway.

↳ How API Gateway decide which MS to call?



Final Diagram:

